

Emanator Theory Using Split Octonions is Manifestly Lorentz Invariant and Reveals Why the Fundamental Constant $\hbar$ Should be so Small

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Abstract

A paper published in 1917 describes Lorentz invariance for complex bi-quaternions [1] (modern notation [2,3]). The chiral bi-sedenion emanator described in [4] is re-examined here with its general octonion, which can be either a regular Cayley octonion or a split octonion (a.k.a., a bi-quaternion), taken to be specifically a split octonion. Thus, we now explicitly consider the bi-quaternion form and show how the formalism retains its desirable properties, explaining $U(1) \times SU(2) \times SU(3)$ for example, while now becoming manifestly Lorentz invariant as well. In adopting the split Cayley algebra form of maximum information emanation zero-divisors occur that are removed to achieve maximal analytic continuation. Removal is an exact process on a complex product space (that is fully split to the chiral noise dimension of 29 ‘free’ real dimensions). The effective dimension will be shown to be slightly greater than 29, however, here called 29*. Knowing an effective dimensionality of a hypothesized maximal information emanation can be used in a number of arguments, including in conjunction with a hypothesis that such maximal information emanation be ‘at the edge of chaos’ – where the fractally optimum Mandelbrot set, with boundary $\dim=2$ [5], is taken as a limit map description for each of the split dimensions that is complexified. The Feigenbaum universal constants [6] occur in the Mandelbrot set, the second of which is denoted here C_∞ , which describes maximum perturbation ‘at

the edge of chaos'. A relation between $\{C_\infty, \alpha, 29^*\}$ is thereby indicated [7]. By considering the proposed maximal analytic continuation from 29^* real dimensions, we arrive at a derivation for an 'h-value' in the formalism that is very, very, small, but with a small dimensionful constant factor it becomes the familiar $\hbar$. Statistical & computational aspects must be re-examined for the split-formalism, as well, but work on that is presented in a separate paper to stay focused on analytic effects here.

Keywords: Lorentz invariant, Trigintaduonion, emanator theory, emanation, the standard model, U(1), SU(2), SU(3), alpha, fine-structure constant, Feigenbaum constant, Planck's constant, quantum matter, path integral, propagation, Cayley algebra

Introduction

A now obscure paper, published in 1917, describes Lorentz invariance for complex bi-quaternions [1] (modern notation [2,3]). In the description of Emanator theory thus far [4,7-12], a general octonion is indicated that can either be a regular Cayley octonion or a split octonion (aka, a bi-quaternion). In most of the analysis thus far the specific form of a Cayley Octonion is assumed, although the first paper [4] refers to "chiral bi-sedenions" in the title. The specific bi-sedenion formulation is what we return to here, because it leads to the manifestly Lorentz invariant form shown in [1]. This is because a split sedenion will be split at all sub-Cayley levels, thus selects the form of the chiral trigintaduonion with a split octonion [13]. The specific results of [4,7-12] 'carry over' with the shift from octonion to split octonion (the prior results typically involve term counting that is independent of sign, and all that changes in the multiplication rule when going from octonions to split-octonions is a single sign). What follows is a review of the above derivations, starting with a background section on the Lorentz transform, followed by background on the prior key results [4,7-12], with their (trivial) modification when shifting from Cayley to split-Cayley described. Notably, the product space argument leading to the explanation for $\hbar$ [11] was previously restricted to chiral emanators that were infinitesimal variations on unity (to allow decoupling at linear order). This restriction is no longer needed in the split formulation as *a full decoupling occurs by construction*. This strengthens the derivation for $\hbar$ significantly so it is placed in the Methods that follows and discussed further in the Results along with the Lorentz Invariant construction.

When examining the chiral trigintaduonion emanation at maximal dimensionality (that would preserve unit norm on the propagated trigintaduonion) we arrived at a template for the 10D chiral form. Recall that a general trigintaduonion T is 32D and can be written in terms of 4 octonions: $T=((a,b),(c,d))$, while the chiral T can be

written in terms on 1 octonion and two real parameters: $T=((a,\alpha),(\beta,0))$, where there are four chiral forms. The choice on the octonion ‘a’ in the chiral form was made in [4,7-12] to stay with the conventional Cayley algebra construction, not the split form. But either choice could be made, and the split form is arguably a larger space (thus a larger channel for maximum information emanation). The original hesitancy to use the split form is it has zero-dividers and other oddities not seen until operating in the full 32D trigintaduonion propagation, which seemed like something to be avoided but later analysis [5] revealed that removal of the ZD’s provides a mechanism for introducing point-like matter. So there is no longer the aversion to having the split form. Furthermore, as indicated, if the ZD structure is not to be avoided, rather maintained, then when working with a chiral-T emanation within the larger T-space, where the larger space permits ZD’s, it would be necessary to choose the split form in the T-chiral space to faithfully carry-over any ZD structure. Note that switching the type of octonion in the chiral form does not alter the independent octonion counting analysis underlying the results of [4,7-12], thus those results remain unchanged.

Background

Lorentz transform notation

We can write the coordinate transformation under Lorentz transform as

$$x^{\mu'} = \Lambda^{\mu'}_{\nu} x^{\nu},$$

where $x^{\mu'}$ is a contravariant coordinate, and if the 4-vector length is invariant, then the metric is invariant:

$$\eta = \Lambda^T \eta \Lambda.$$

Any 4-vector entity z^{ν} that transforms as x^{ν} , is, similarly, called contravariant. This is to distinguish it from covariant 4-vectors that transform as:

$$z'_{\mu} = (\Lambda^{-1})^{\nu}_{\mu} z_{\nu}.$$

Note that for rotation transformations in 3D we have $R^{-1} = R^T$, so the ‘3-vectors’ transform identically, and there is no differentiation of contravariant and covariant vectors. For the 4D Lorentz transform, however:

$$\eta = \Lambda^T \eta \Lambda \rightarrow \Lambda^{-1} = \eta^{-1} \Lambda^T \eta$$

and Λ^{-1} and Λ^T are not the same.

The 4-vector reformulation of EM – proof of Lorentz Invariance by construction [14]

If the 4-vector length is invariant under Lorentz transform, then so is the metric as indicated previously. Taken further, this invariance extends to any scalar or scalar product. The general transform properties for a full tensor generalization then follow. To show that the Maxwell equations describe a Lorentz invariant theory, we must identify 4-vectors that might be formed from the electric and magnetic fields. Such 4-vectors, if grouped (inner product contraction) to form a scalar entity, will be

manifestly Lorentz invariant. If such scalar entities can be constructed that coincide with the Maxwell equations, then the proof of Lorentz invariance is complete by construction. Recall the component notation for the E and B fields in terms of potentials:

$$E_k = -\nabla_k \varphi - \frac{\partial A_k}{\partial t} \quad \text{and} \quad \epsilon_{ijk} B^k = \partial_i A_j - \partial_j A_i$$

So far we have the 3-potential A_k and need to generalize it to a ‘4-potential’. Furthermore, the 3-gradient operator has a natural generalization to a 4-gradient operator (that transforms like a covariant vector). If we identify the 4-potential to be:

$$A_\mu = \left(-\frac{\varphi}{c}, A_k \right),$$

then (setting $c=1$):

$$E_k = \partial_k A_0 - \partial_0 A_k.$$

The forms E_k and $\epsilon_{ijk} B^k$ suggest that we examine the antisymmetric 2nd rank tensor formed from the 4-potential by the relation:

$$F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha,$$

where we see the E and B components reside in this structure:

$$F_{\alpha\beta} = \begin{bmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & B^3 & B^2 \\ E^2 & -B^3 & 0 & B^1 \\ E^3 & B^2 & -B^1 & 0 \end{bmatrix}$$

Let’s consider a Lorentz transform corresponding to a boost in the x-direction (and no longer have $c=1$), then $\Lambda^0_0 = \gamma = \Lambda^1_1$, and $\Lambda^0_1 = -\gamma v/c = \Lambda^1_0$, and $\Lambda^2_2 = 1 = \Lambda^3_3$, with the rest zero. Let’s see how the E and B components transform:

$$\begin{aligned} E'^1 &= E^1 \quad \text{and} \quad B'^1 = B^1 \\ E'^2 &= \gamma(E^2 - vB^3) = \gamma(E^2 + (v \times B)^2) \\ &\quad \text{etc.} \end{aligned}$$

These are indeed the transformations consistent with the coordinate transform of Maxwell’s equations (can be shown by considering an infinitesimal Lorentz boost as indicated). Since we began with a potential formulation, this is then equivalent, without loss of generality, to a proof of Lorentz invariance for the homogenous Maxwell equations. For the inhomogeneous Maxwell equations we have

$$\rho = \epsilon \nabla \cdot E = \epsilon \partial_k F^{0k} \quad \text{and} \quad J^i = \epsilon \partial_k F^{ik} + \epsilon \partial_0 F^{i0}$$

thus we form the 4-vector for current to be:

$$J^\mu = (\rho, J^i),$$

which means that the inhomogeneous Maxwell equations can be written as a contravariant 4-vector expression (which is manifestly Lorentz invariant):

$$\epsilon \partial_\nu F^{\mu\nu} = \frac{1}{c} J^\mu.$$

Lastly, for the effect of an electromagnetic field on a moving particle we have the Lorentz equation:

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \mathbf{p} = \gamma m \mathbf{v}.$$

To get to a 4-vector formulation we already know that $\mathbf{p}$ generalizes to the 4-vector:

$$p^\mu = (E/c, \mathbf{p}).$$

On the RHS we have a $\mathbf{v} \times \mathbf{B}$ term. This suggests a tensor form involving $F^{\mu\nu}$ (to get the B^k term) and the velocity 4-vector $U^\mu = (\gamma c, \gamma \mathbf{v})$, for which we find:

$$q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \rightarrow \frac{q}{\gamma c} F_{j\mu} U^\mu$$

The 4-vector form of the Lorentz equation then becomes:

$$\frac{dp^\mu}{dt} = \frac{q}{\gamma c} F_{\nu\mu} U^\nu.$$

which is manifestly Lorentz invariant. Thus the Maxwell equations and the coupling of the EM Fields to charged matter are all Lorentz invariant. (Note that charge is a scalar, thus is seen to be the same in all Lorentz frames.)

Appearance of Lorentz-invariant Spinor Solutions in addition to Vector solutions [15]

A mathematical subtlety that arises at this juncture is not due to the novel three boosts introduced by the 4D Lorentz transformation, but due to the three (standard) rotations. This is because the rotation group in 3D, $SO(3)$, admits a double cover from the group $SU(2)$, indicating two types of solutions. We find that there is a new type of Lorentz invariant description, other than the 4-vectors, known as a spinor. The easiest way to show that Lorentz invariance extends to the new mathematical object is to rewrite a 4-vector as a 2x2 Hermitian matrix, where V^a with components (V^0, V^1, V^2, V^3) is written as the 1-1 mapping as

$$\psi(V^a) = V^{AA'} = \begin{pmatrix} V^{00'} & V^{01'} \\ V^{10'} & V^{11'} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} V^0 + V^3 & V^1 + iV^2 \\ V^1 - iV^2 & V^0 - V^3 \end{pmatrix}$$

where the length of the 4-vector is related to the determinant of the matrix:

$$\det[\psi(V^a)] = \frac{1}{2} \eta_{ab} V^a V^b.$$

Thus, there is a map from $SL(2, \mathbb{C})$ to the Lorentz transforms where there is a 2-1 isomorphism. This can be seen to be directly related to the $SU(2)$ double cover (2-1 mapping) on $SO(3)$ as mentioned previously (we consider the Lorentz transform leaving the time-like component of a chosen orthonormal tetrad invariant, what remains is the 3D rotation group $SO(3)$). This is not simply a useful mathematical relation. We've identified the invariance under Lorentz transform as a fundamental element of the theory (giving rise to special relativity, etc.). *If we see this then conveyed to an extended set of spinorial solutions, in addition to vectorial, then new*

forms of energy/matter are indicated for the spinorial solutions. This is precisely what is observed: matter is spinorial (spin $\pm 1/2$) and (force) fields are vectorial.

Lorentz invariance via complex biquaternions [1-3]

Let's now consider a similar process involving transformational invariance but instead of encoding the Lorentz transform in the form of matrix transformation invariance let's use elements of the Cayley algebras instead. Specifically, consider the following transformation:

$$q' = aqa_c^*, \text{ where } aa_c = 1,$$

where a is a (unitary) complex bi-quaternion: $H(\mathbb{C}) \times H(\mathbb{C})$, and $q = (ct, ix, iy, iz)$ (note this notation is for $q = (ReH_1, ilmH_1, iReH_2, ilmH_2)$). The $q' = (ct', ix', iy', iz')$ that results will correspond to a proper orthochronous Lorentz transform [1-3].

This is a remarkable result, but is there better (higher dimensionality/complexity)? Consider that a complex bi-quaternion is isomorphic to a complex octonion which is isomorphic to a 'chiral' sedenion, thus a theory for chiral sedenion emanation is indicated from this result as far back as 1917. The 'halting condition' on the generalization in 1917 seems to be that octonions are the highest-order of the division algebras that have inverses defined (which is necessary to have $aa_c = 1$ be defined). But they have already extended past octonions since these are complex octonions $\cong$ chiral sedenions, so how are they guaranteed to have $aa_c = 1$ be defined? This is possible for the *chiral* sedenions, as shown in [4], if restricted to be unit norm. So now we have our answer based on the results from [4] – we can go one complexation order higher:

$$Q' = AQA_c^*, \text{ where } AA_c = 1,$$

where A is a (unitary) bi-complex bi-quaternion: $H(\mathbb{C} \times \mathbb{C}) \times H(\mathbb{C} \times \mathbb{C})$, which is isomorphic to a unitary quaternionic bi-quaternion, $H(\mathbb{H}) \times H(\mathbb{H})$, and $Q = (ct, ix, iy, iz)$.

where $Q = (ReH_1, ilmH_1, iReH_2, ilmH_2)$ as before, except $H_1 = H \times \mathbb{H}$ not $H_1 = H \times \mathbb{C}$

The $Q' = (ct', ix', iy', iz')$ that results will again correspond to a proper orthochronous Lorentz transform. Again, how do we know the critical operation $AA_c = 1$ can always be satisfied? Previously we saw that a complex bi-quaternion was equivalent to a 'chiral' sedenion (in the sense described in [4]), thus a bi-complex bi-quaternion is isomorphic to a complex chiral sedenion, which is isomorphic to a 'doubly chiral' trigintaduonion. This is precisely the construct examined in [4], so a generalization of the 1917 result to unitary quaternionic bi-quaternions appears possible. As shown in [4], however, there is no higher order construct. This latter form (on doubly-chiral trigintaduonions) establishes the Emanator as Lorentz Invariant explicitly. To consider the 1917 relation (involving

chiral sedenions) directly in the analysis, we shall simply expand the doubly chiral trigintaduonion to a (singly) chiral sedenion and operate at that level in the analysis.

Existence of generalized unit-norm propagation structure that is 10D [4]

Unit-norm right product propagation is trivial for the division algebras since $\text{norm}(XY) = \text{norm}(X) \times \text{norm}(Y)$. From this it is apparent that we have an automorphism group given by the norm itself ($A(XY)=A(X)A(Y)$), and in the case of the octonions this automorphism group is G_2 [16]. It can be shown that $SU(3)$ is in G_2 [16]. Let's now consider the situation with a higher-order Cayley algebra, the Sedenions, 'S'. We obviously don't have $\text{norm}(S_1 S_2) = \text{norm}(S_1) \times \text{norm}(S_2)$ in general, as this would then allow S to join the ranks of the division algebras, and it is proven that such don't exist above the Octonions [17]. Can we still have a propagation structure? Is it possible to have a 'base' sedenion for which $\text{norm}(S_{\text{base}})=1$, and to have a right propagator (product) sedenion also $\text{norm}(S_{\text{right}})=1$, such that $\text{norm}(S_{\text{base}} \times S_{\text{right}}) = 1$? The answer is yes (see appendix of [12] and [4]), when the sedenion has the (chiral) form of an octonion crossed with a real octonion: $S_{\text{chiral}} = (O, O_{\text{real}})$ or $S_{\text{chiral}} = (O_{\text{real}}, O)$. Can we continue this to arrive at a propagation structure on the Trigintaduonions? Again the answer is yes, with the chiral form generalizing off the chiral Sedenion as might be expected: $T_{\text{chiral}} = (S_{\text{chiral}}, S_{\text{real}})$ or $(S_{\text{real}}, S_{\text{chiral}})$ [4]. It is proven that this extension process will go no further [4]. What happens is that due to the chiral form we are still able to re-express all T products (or S) as collections of terms involving tri-octonionic products (which have nice properties as described in [4]), and this can no longer occur above the (chiral) trigintaduonion level.

Chiral Trigintaduonion emanation involves 137 independent octonionic terms

This is too lengthy to conveniently repeat in the background here, see [12].

Achiral Trigintaduonion unit-norm propagation has maximum perturbation $\propto$ [8,9]

The maximum perturbation, referred to as maximum noise in what follows, is first evaluated for a chiral emanation where we take a $\text{norm}=1$ $T_{\text{base}}=(A,B)$ and take the right product with T_{chiral} in the form $T_{\text{chiral}}=(C,\beta)$, with product $(A,B)(C,\beta)$ proven to be unit norm above [4]. In the prior section we saw that there are 137 independent octonion terms at the octonion sub-level of the new unit norm trigintaduonion that results, which leads to 137 independent terms at component level. In order to use the map rules mentioned in the previous section, it is necessary to move from the trigintaduonion, T, space, to the $C \times T$ space. This is done in later sections where we consider sums on $\exp(iT)$. The exponential function (map) provides a well-defined 'lift' of a hypercomplex (Cayley) algebra from T to $C \times T$. The exponential map also provides a very useful maneuver when working with unit-norm hypercomplex

numbers via the generalized deMoivre theorem $\exp(\text{Im}(T)) = \cos(\theta) + \text{Im}(T)\sin(\theta)$, with the real part recoverable from $\cos(\theta)$. More details on this follow later but for now, in evaluating the maximum noise allowed we have three structures to adopt: (1) the noise is generalized to be complex (as will be the case for the components themselves once the $T \rightarrow C \times T$ structure is adopted). (2) At component-level, the noise (for maximum noise) is equipartitioned in both real and imaginary parts. (3) Total imaginary noise magnitude is π for maximal antiphase (as described previously).

(I) Chiral emanation noise: have 137 terms with max unit norm each, for the real part, and for the imaginary part have a “phase angle” β such that $137\beta = \pi$ (here referred to as a phase angle in the sense that the $\exp(\text{Im}(T))$ map is being used). The noise magnitude at octonionic-component level is then given by the right triangle with real part = 137 and angle $\beta = \pi/137$, thus maximum chiral emanation noise magnitude is:

$$H = 137/\cos(\pi/137)$$

(II) Achiral emanation noise: now have 29 “free” components, each with 137 independent terms. For maximum achiral emanation we thus have 137×29 independent terms that are built from the aforementioned chiral emanation terms (to make achiral). If we equipartition as before, with noise magnitude H_c , we have a “noise triangle” with magnitude (hypotenuse) H_c and with angle $\theta = \pi/137 \times 29$. The imaginary part is then $(H_c)\sin(\theta)$. As regards the H magnitude separated form (separating out the ‘H’ factor for now), we have for the imaginary part $\sin(\theta)c$. As before, we take maximal noise transmission when all the imaginary parts add to maximal antiphase. Given the equipartitioning assumption, we then simply have the factor 137×29 :

$$\sin(\theta) c (137 \times 29) = \pi \rightarrow c = \theta / \sin \theta.$$

The maximum real noise perturbation that the system can have is then α , where:

$$\alpha^{-1} = \frac{137}{\cos \beta} \cos \theta \frac{\theta}{\sin \theta}, \quad \text{where } \beta = \frac{\pi}{137} \text{ and } \theta = \frac{\pi}{137 \times 29}$$

$$\alpha^{-1} = 137.03599978669910,$$

where the evaluation was done at WolframAlpha to high precision [18] (e.g., higher precision than that reported in earlier work [8]). This matches the experimentally observed value to all 11 decimal places currently known. As of 2002 [19], the measured value of α is:

$$\alpha^{-1} = 137.03599976(50).$$

Note that in quantum field theory the parameters are renormalized at a particular energy scale. Thus choice of energy scale impacts the value of α (as a coupling constant in the classical theory or a perturbation expansion factor in the quantum theory). At 0K we have the extreme low-energy end of the renormalization group

(with the largest α value). We are at the 2.7K CMBR, so we have the max α to very high precision. (In studies at high energy scale at LEP, at the energy scale of the Z-boson (91GeV), we get the renormalized value to be [19,20]: $\alpha^{-1}[M_Z] \cong 127.5$. Note that 91GeV is way above the energy scale of the familiar Hagedorn temperature at $\sim$ pion mass=150MeV or 1.7×10^{12} K) [21], where hadronic matter ‘evaporates’ into quark matter.)

Achiral emanation using a 72-card deck

There are 4 chiralities, so to get an achiral emanator candidate, minimally need a “4-card deck” to emanate in the four chiralities, with emanator equal to normalized sum. The actual deck appears to require a normalized sum over sub-chiralities, as will be explicitly enumerated in what follows. Regardless of the form of the achiral sum over chiral variants, since each chiral emanation has 29 free components, their norm sum will again have 29 free components. Thus, the form for α^{-1} shown above is the complete, non-approximating, result since it only need rely on 29 free component number. Further analysis of the emanation deck, beyond ‘dealing a single card’ is in [12].

Single-step achiral 72-deck emanation has noise propagation dimension $\cong 29^*$ [12]

Obtaining an achiral emanation from a collection of chiral emanations requires that all chiralities be summed over (there are four) as well as sub-chiralities (there are 72). Noise analysis requires collecting of first-order terms. Analysis of noise transmission (when dealt a ‘single-card hand’) indicates 29^* dimensions, where:

$$29^* \cong 29 + \left(\frac{4\pi}{72}\right) \left[1 + \left(\frac{\pi}{137 \cdot 29}\right) \left(\left(\frac{\pi}{72}\right) + \left(\frac{3}{72}\right) \right) \right]$$

The above result was obtained in [7] to describe the 72-card chiral ‘deck’ of chiral emanation products for a single-step emanation. In [12] this is reviewed and elaborated further.

‘Edge of chaos’ maximal perturbation hypothesis [7]

Consider the ‘edge of chaos’ maximal perturbation in each of the 29^* dimensions to be at position C_∞ (see Appendix in [X] for relevant background on Mandelbrot Set), which is on the negative real axis, i.e., at π rotation to have -1 factor, **thus at maximal antiphase**. This results in the relation for maximal perturbation at maximal antiphase (maximum reference angle with sign chosen positive by convention) has a lower bound on α given by:

$$\alpha_0^{-1} = \left(\sqrt{C_\infty}\right)^{29^*}$$

where

$$C_{\infty} = 1.4011551890920506004 \dots$$

This ties $1/\alpha$ to the second Feigenbaum constant C_{∞} in the context of the Mandelbrot set. It is well known that the Feigenbaum constants are universal, and part of a description of a universal transition to chaos regime. The Mandelbrot set is also universal [22], and maximal in that its fractal boundary has maximal fractal dimension of 2 [22], a detail that will be important in the meromorphic matter description given later.

For C_{∞} , most references only provide $C_{\infty} = 1.401155189 \dots$, and a higher precision tabulation is not readily found, so use is made of the relation

$$C_n = a_n(a_n - 2)/4,$$

together with the tabulation on a_{∞} [23]:

$$a_{\infty} = 3.5699456718709449018 \dots$$

The resulting C_{∞} is:

$$C_{\infty} = 1.4011551890920506004 \dots$$

The resulting α_0^{-1} is:

$$\alpha_0^{-1} = 137.03599933370198263 \dots$$

For the multi-card analysis we have:

$$\alpha^{-1} = \alpha_0^{-1} + \alpha_1^{-1} + \dots$$

where α_0^{-1} involves the sum over emanation by one-card. For sum on two-chiral products we have further ‘noise’ contribution α_1^{-1} . With the multi-card modifications, albeit small, there is the complication of shift from 72-deck to 78-deck, and whether there is a chiral step (‘card’) type that can be exactly repeated (i.e., are cards from the deck played with replacement when considering a multi-card flop sequence). There may be a reason why the sums must be done without card-replacement. This might be because card replacement would allow degenerate tri-octonionic product terms, again throwing off the 137 braid term total, perhaps, leading to non-optimality. This is being explored in further work and will not be discussed further at this time.

Chiral trigintaduonion emanation leads to the standard model of particle physics and to quantum matter

The chiral trigintaduonions T , with right product operation $((T \times T) \times T) \dots$, used previously for maximum information transmission, are here shown to be $H \times O$ when arranged for achiral emanation. When considering a sum over chiral emanations to obtain an achiral emanator, with T as phase factor, we have the exponentiation operation $\exp(iT)$, which leads to a theory that is $C \times H \times O$. As such, we have the foundation for the associative operator algebra of the Standard Model: $U(1) \times SU(2)_L \times SU(3)$. A complication with T products is you can have zero divisors. A framework

is adopted to remove the zero divisors by requirement of maximum domain of analyticity on the log trigintaduonion multiplication, resulting in a description for the meromorphic precipitation of matter. In this process a fundamental quantum is indicated from the zero-divisor residue terms. Analyticity in the form of a Wick rotation also provides a mechanism whereby we can transition to a dimensionful action and quantum and arrive at an explanation for the critical ‘smallness’ of Planck’s constant. The emanator is $T_{em} \cong H \times O$, and provides a description of a possible meromorphic origin for point-like matter.

Getting an associative algebra from the repeated operation of a non-associative algebra is first described in [16] in the context of repeated octonion products: $((O \times O) \times O) \dots$, where the algebra $SU(3)$ can result. This is found to be equivalent to fixing one of the octonion imaginary components in such a repeated-product operation [16,24]. Dixon shows in [24] that the $C \times H \times O$ product algebra lays the foundation for the associative operator algebra of the Standard Model: $U(1) \times SU(2)_L \times SU(3)$. In later work this is explored in the form of ideals [25]. Emanator theory, described in a brief background to follow, is based on unit-norm propagation at maximal dimension. It turns out this maximal dimension is not octonion-based but trigintaduonion-based [4], although it does have an octonion sub-algebra: $T_{em} \cong H \times O$, as will be shown here.

In prior work [4,7-12] we hypothesized maximal algebraic information flow, where the “emanation” of information is represented as multiplication by an element of an algebra in two steps: (i) take the maximal current-state element that is a unit-norm trigintaduonion; and (ii) perform the emanator step that consists of an achiral sum of multiplications with chiral trigintaduonion emanators. In prior efforts this was considered without the complication of zero divisors. We will see that zero divisors act as “sources”, so the prior work was effectively analysis of sourceless information flow.

In [12] we show $T_{em} \cong H \times O$ (which will lead to the standard model) and we consider zero divisors and their impact on the maximal information flow and in doing so see a mechanism for meromorphic precipitation of quantum matter with dimensionful action. When we consider

$\exp(iT_{em})$, we get a mathematical object that is $C \times H \times O$, thus whose product algebra becomes $U(1) \times SU(2) \times SU(3)$ according to Dixon [24]. If this route to the standard model is unsatisfactory, consider that the T emanation sum pre-normalization can be grouped as a 2×2 real matrix that is finite, thus $SU(2)$ with normalization: $T_{em} \cong [2 \times 2 \text{ real}] \times O$;

where repeated steps of
 $\exp(iT_{em}) \rightarrow \text{unit norm } C \times [2 \times 2 \text{ real}] \times O$

re-group as repeated steps of $CxCxCx\dots$, and $[2 \times 2 \text{ real}] \times [2 \times 2 \text{ real}] \times \dots$, and $OxOxO\dots$, with unit normalization, thus the product algebra is $U(1) \times SU(2) \times SU(3)$.

Why 22 parameters?

The Trigintaduonion emanation hypothesis strongly indicated the possibility of only 22 parameters in early efforts [9]. In [12] we see a clearer mathematical argument for this being the case. Consider the emanator for the excluded case of a pure real common octonion δ :

$$T_{chiral}^{(k)} = \begin{cases} ((\delta, \alpha), \beta) \\ ((\alpha, \delta), \beta) \\ (\beta, (\delta, \alpha)) \\ (\beta, (\alpha, \delta)) \end{cases}$$

We get no sub-octonionic mixing for this type of emanation: $\mathbf{T}_{em} = \mathbf{O} \times \mathbf{H}$, where $\mathbf{O} = \mathbf{1}$. Such an emanation, acting on T_{base} , in order to remain achiral, will change nothing vis-a-vis future achiral T_{em} products, thus it is associated with a “constant of the emanation”. How many such emanations, effecting no change, are there? They correspond to the non-octonion based fluctuations in the evaluation of the 78 deck described in [7,12], of which there are 22.

Methods

Maximal emanator analyticity via removal of zeros-divisors

ZD-removal was originally done on a hypothesized $[\mathbb{R} \otimes]^{29}$ decomposition but this decomposition is now accomplished without approximation, by construction, by use of the split-Cayley form. The $[\mathbb{R} \otimes]^{29}$ is then taken to be real part of a $[\mathbb{C} \otimes]^{29}$ analytic continuation, where a double-pole residue [12] leads to a derivation for a ‘1/big’ QM-factor to be discussed later. The decomposition (via split) then provides a complex extendable noise propagation effective dimensionality 29*, thus have derivation of ‘h*’ in [12]. The presumption was that ZD-encounters could occur in the (max) perturbed 10D chiral-T propagation since it is probing the 32D T-space where ZD’s are known to exist. With the split-formulation, there are ZD encounters for the 10D chiral-T propagation even with no perturbation. This will be more suitable for the ZD surgery argument since now instead of having to be infinitesimally near a chiral-T trajectory, we can simply take the chiral-T trajectory.

The general form of maximal, analytic, information emanation [12] gives rise to a sum on residue-like terms associated with each of the zero-divisors (zd’s), and an ‘action’ variable is indicated to result from the parameterization of the approach to each of the zero-divisors, with their individual actions additive (phase contributions

multiplicative) for parts contributing to a particular zd. The sum over all the zd's will, upon analytic continuation, be associated with a sum over paths. The zd action variable is written in the form of the integral of a functional along a path parameterized by 'time', with the usual definition for Action if the functional is the Lagrangian:

$$\sum_{\text{zeros } S^*_i} e^{S^*_i/|h^*|} \rightarrow \int_{\text{matter paths}} e^{iS(i)/h}, \text{ where } S(i) = \int L dt,$$

where the definition for action above is kept to the simple form for a point particle trajectory. More complex forms can be written for field descriptions, where we generalize from point particle forms in various ways, but still with point-like coupling terms. Further generalization to actions describing 1-D objects, string not points, or beyond (n-D objects, or branes) and their trajectories is possible at this point but note how the chain of associations is altered, if not broken. Tracing back to a fundamental issue of analyticity when going from emanator form to propagator form, we saw that analyticity requires *isolated* zeros to not make the entire solution trivially zero. Thus, the fundamental meromorphic 'precipitate' for matter might be point and point-based field constructs as we've developed them already, where the role of string theory emerges separately (perhaps on space-time boundaries), although possibly as early as the "matter-precipitating" Wick rotation step above. Note that in going from

$$\frac{S^*_i}{|h^*|} \rightarrow i \frac{S(i)}{h}$$

(1) Both ratios are dimensionless, but the quantities on the left are a ratio of pure numbers themselves dimensionless, while the RHS has a ratio of S, the action, with action dimensions, and h, Planck's constant, also with action unit dimensions.

(2) Both $1/|h^*|$ and " $1/|h|$ ", where the abs value operation on the latter simply drops the dimensionful units, are extremely large numbers and occurring in a phase argument. This *sets up a highly oscillatory integral* (see Appendix in [11]) such that the classical solution $\delta S = 0$ results (if a classical solution exists for the system studied), among other things.

Results

Fundamental emanator hypothesis: maximal information propagation $\rightarrow$ Lorentz-Invariant point-like matter

Maximum information flow occurs for octonions when split, because additional structure can occur, such as zero divisors. Maximum information flow with a propagate-able mathematical structure is thus a chiral trigintaduonion of the form

$$T_{chiral}^{(k)} = \begin{cases} ((O_{split}, \alpha), \beta) \\ ((\alpha, O_{split}), \beta) \\ (\beta, (O_{split}, \alpha)) \\ (\beta, (\alpha, O_{split})) \end{cases}$$

Point-like matter appears in the theory when the zero-divisors in the propagation are managed by removal according to maximal analytic domain (the maximum flow hypothesis will occur in a maximal domain). Thus, it is found that the zero-divisor general formulation of the maximal propagator is indicated by maximum information flow and indicates point-like matter. This formulation involves the split-Cayley algebras. Thus, the chiral trigintaduonion is a chiral bi-sedenion as originally indicated [4]. The chirality at sedenion-level is:

$$S_{split,L} = (O_{split}, O_{real}) \text{ or } S_{split,R} = (O_{real}, O_{split}).$$

If we are working with a chain of products of chiral emanator trigintaduonions $T_{chiral}^{(1)}$ this will reduce to a chain of products of $((O_{split}, \alpha), \beta)$ and upon using the unit-norm constraint to eliminate β as an independent variable, we have a product of (O_{split}', α') . So, dropping the primes, we have products:

$$(O_{split}, \alpha)_1 \times (O_{split}, \alpha)_2 \times \dots$$

Since $O_{split} \cong H \times H$ and $(O, \alpha) \cong O(\mathbb{C})$, we have

$$(H(\mathbb{C}), H(\mathbb{C}))_1 \times (H(\mathbb{C}), H(\mathbb{C}))_2 \times \dots$$

Recall in the background section that if we wanted to consider the standard equivalence class of invariants for unitary complex bi-quaternion operations, then we call

$$(H(\mathbb{C}), H(\mathbb{C}))_1 = a_c^*, \quad aa_c = 1, \quad |a| = 1,$$

and have propagation according to:

$$qa_c^* \rightarrow q_{new}$$

and invariance according to

$$q' = aqa_c^*$$

where invariance for $q = (ct, ix, iy, iz)$ is realized such that the $q' = (ct', ix', iy', iz')$ that results will correspond to a proper orthochronous Lorentz transform. The maximal information hypothesis underlying the proposed Emanator theory, thus indicates Lorentz invariant point-like matter.

The maximal noise propagation dimension hypothesis

$$\alpha^{-1} = \left(\sqrt{C_\infty}\right)^{29^*}, \quad \text{where } \dim. = 29^*.$$

The ‘smallness’ of Plank’s constant explained

We now consider zero-divisor (zd) removal for a hypercomplex function in the 29^* dimensional noise propagation space. If the emanator is to remain an achiral mix, as well as analytic, then we can't allow the zero divisor events that would drop a chirality mentioned above, where these occurrences are treated as isolated singular events removable from the domain of definition of the emanator by repeated application of the analytic domain 'surgery' (repeated on both events, and for given event, at component level). This analytic 'surgery' occurs for each of the independent component dimensions for a given chiral emanation, and for each of those dimensions it returns a zero-removal 'residue' of $4\pi i$ (with an extra factor of 2 since a double log zero at the fractal boundary). We found in [7] what the effective dimension is 29^* , thus the remnant of the surgery for each zd removed is:

$$\frac{1}{|h^*|} = (4\pi)^{29^*}.$$

In the above, we assumed a $(\mathbb{R} \times)^{29^*}$ local factorization existed and previously considered explicit cases with chiral T's that were infinitesimal variations on unity in order to ensure decoupling of terms (at first order) to effect a $(\mathbb{R} \times)^{29}$ factorization. None of this constraint is needed in the split formulation, where we trivially have the factorization due to the split basis. We then analytically extend to being the real part of a $(\mathbb{C} \times)^{29}$ that would then be extended to the maximal (fractal) dimension 29^* : thus $(\mathbb{C} \times)^{29^*}$. We have:

$$|h^*| = 6.630 \times 10^{-33}$$

compare this with Planck's constant $h = 6.626070 \times 10^{-34} \text{ J s}$, where the two 'h' numbers satisfy the extreme smallness property, thus providing integrals with large parameter and thus *a highly oscillatory integral with stationary phase domination, the foundation of the quantum mechanics path integral formulation.*

Discussion & Conclusion

Maximum information hypothesis and Black Holes in Emanator theory

If matter is point-like consisting of a zero-divisor removed for maximal analyticity, then the matter at a BH horizon can be seen as an accrual of removals for maximal analyticity that encompasses the horizon boundary. This can be seen as the boundary being 'removed' or matched with some appropriate boundary condition. Such as the thermally matching boundary condition indicated for the 'fuzzball' string theory solution [26]. Note that the 10D chiral emanations that projects to the Black Hole horizon may relate to the 10D construct of Maldacena [27] (5D Anti-deSitter geometry and 5D conformal field theory) that relates to the 10D string theory construct of Vafa [28]. Thus, there are many hints of connections between the theories, with string theory providing the needed BH boundary solution (and likely interpretation: instead of 'gore at the singularity' there is 'shattering at the horizon').

22 parameters

Emanator theory predicts 22 constants, which would occur if RH sterile massive neutrinos existed, in an extended Standard Model. The observation of neutrino oscillations revealed that the LH neutrinos are massive [29], while recent measurements of $g-2$ for the muon [30] suggest that there are RH massive neutrinos, so this may complete the theory. Note that mass relations are also indicated, where such masses are related, as indicated by the Koide relation [40], with family group formulation to explain this according to [41].

Universal Thermality

There is a fundamental complex structure that still remains from the Emanator formalism upon projection from the higher-dimensional Cayley algebras into the maximum propagation described, for the alpha maximum-perturbation trigintaduonion. That complex structure is ‘contact’ analytic and is realized under conditions where limits involving that structure are taken to zero (or to some fixed value at component-level). Two main examples where this has been most significant are in (i) shifting a QFT to a thermal QFT by shifting to imaginary time related to the inverse temperature [31]; and (ii) in the use of dimensional regularization to renormalize QED and QCD [32]. Thus, the process of emanator theory settling on the maximum information propagation dimension is one where higher order complex structure (from non-propagate-able dimensions) is still accessible for regularization processes. A possible *origin* for the complex structures is described in the next section.

The significance of the remnant complex structure is **universal thermality**. Thermality via complex structure and Lorentz Invariance on chiral Trigintaduonion are more fundamental constructs than QFT (based on the Standard Model) or GR (with manifold dynamics according to the Standard Cosmological Model). This is because the limits defining such thermality would exist for a single chiral Trigintaduonion propagation step, and we don’t ‘see’ QFT until enough emanation steps have occurred such that the product group gauge that begins to resolve is the gauge group of the standard model, thus giving rise to the QFT of the Standard Model.

Universal thermality explains the odd behavior seen upon ‘complexification’, such as BH geometry complexification giving the Hawking temperature of a Black Hole [33] consistent with the full quantum field theory analysis at the BH horizon demonstrating thermal emission with that same temperature done by Hawking [34]. Given the Hawking radiation, a BH is clearly unstable and will evaporate over time. If the BH is ‘placed in a box’ or has not-flat asymptotic geometry, however, then the ‘closed’ system may reach an equilibrium and be stable. This is precisely what is considered by Hawking and Page (1983) [35] upon examining the thermodynamics of

AdS BH's. Further work along these lines was done in [36-38]. In these efforts a Hamiltonian formulation for Gravity is made and then complexification yields a partition function describing an ensemble of BH solutions. Stability, in the form of positive heat capacity, can then be shown.

Hypothesis for Doubly Chiral Split Octonion Emanation with a Doubly Complex Structure

The doubling limits on extension, as well as the extensions themselves, are artefacts of the Cayley algebra construction that is employed. The original double-chiral extension was done to extend the unit norm propagation property from octonions to the trigintaduonions [4]. Here is the emanation expression in various forms:

$$T_{base,new} = T_{Em}\{T_{base,old}\} = \sum T_{base,old} T_{chiral}^{(k)} = T_{base,old} \cdot T_{Em,Eff}.$$

In the last form we speak of an effective emanation ($T_{Em,Eff}$) off of the original base trigintaduonion. In the explicit sum form we see that emanation is a sum over chiral multiplications, where a chiral trigintaduonion has the distinctive ‘doubling’ from of extension:

$$T_{chiral}^{(k)} = \begin{cases} ((O_{split}, \alpha), \beta) \\ ((\alpha, O_{split}), \beta) \\ (\beta, (O_{split}, \alpha)) \\ (\beta, (\alpha, O_{split})) \end{cases}$$

Consider the first chiral emanator form: $((O_{split}, \alpha), \beta)$. In [4] we began with unit-norm propagation on octonions:

$$norm(O_{base} O_{propagator}) = 1 \text{ if } norm(O_{base}) = 1 \text{ and } norm(O_{propagator}) = 1$$

and extended via the Cayley hierarchy two level higher to the trigintaduonions:

$$norm(T_{base} \cdot T_{Em,Eff}) = 1 \text{ if } norm(T_{base}) = 1 \text{ and } norm(T_{Em,Eff}) = 1.$$

By going two-level higher we can still expand the Cayley products two-levels down, back to the level of tri-octonionic products, where the norm operation is defined and associative in the sense that $norm(O_1 O_2 O_3) = norm(O_1) norm(O_2) norm(O_3)$ (true for division algebras such as the octonions).

What is proposed now is a similar extension to that of $O \rightarrow ((O, \alpha), \beta)$, where the norm property (and associativity under norm) was extended. Here we want to extend:

$$T_{Em,Eff} \rightarrow ((T_{Em,Eff}, \gamma), \delta).$$

Before we were trying to extend a norm property with implicit norm associativity on that operation, here we merely want to extend the theory in a well-defined way that captures “maximal information emanation”. At this juncture it would be nice to assume or impose a selection process on the product of effective emanations that is explicitly associative:

$T_{base,old} \cdot T_{Em,Eff1} \cdot T_{Em,Eff2} \cdot T_{Em,Eff3} = T_{base,old} \cdot T_{Em,Eff2} \cdot T_{Em,Eff3} \cdot T_{Em,Eff1}$
 but to do so would clearly be limiting on the “maximal information flow” by dropping the non-associativity (think of it as dropping 1D of information). In the context given, however, we are extending via two layers of complex structure indicated by the parameters γ and δ (*so extending by 2D for a net gain of 1D*). Thus, we hypothesize that maximal information flow will select effective chiral emanations to be associative such that two layers of complex structure can be carried by the maximum perturbation emanation (where $\gamma \rightarrow 0$ and $\delta \rightarrow 0$ to then give the ‘contact’ complex structure indicated above. The selection for associativity may dictate the ‘size of the hand dealt’ in the chiral emanator deck discussion, where the number of cards, e.g. a sum of chiral emanation steps (78 or 72), are ‘dealt’ until associativity vis-vis the preceding two achiral (summed) emanations steps is achieved. Further elaboration on these ideas will be made in a later paper.

Apparatus, Measurement Theory, and related Emanation Processes

In work involving a full general relativistic solution to describe dust shell collapse it was found that the geometry is effectively “pure apparatus” [39]. Geometry is thus at least part of the apparatus. Consideration of ‘bare’ QFT in an arbitrary apparatus leads to point like violation of the weak energy condition (WEC), but with choice of apparatus can have average WEC (AWEC) such that no violation of AWEC is seen. The separation of quantum observable and classical apparatus has always been a troublesome complication; but in Emanator theory this is simply seen as a fundamental aspect of the emanation process. Consider repeated ‘emanation’ steps starting from Tbase, we then have two aspects: Tbase vs the product algebra (Tbase x Tem) x Tem x ... Consider these objects:

Tbase: this is a higher-order spinor, spinor fields describe (quantum) matter
 (Tbase x Tem) x Tem x ... : gives the standard model gauge field that is Lorentz Invariant

From the Lorentz Invariance on 4D we get local Minkowski space-time, which then generalizes to a manifold globally. So, geometry appears in this process in the context of the product algebra (taken to be equivalent to apparatus) and the matter appears in spinor form (taken to be a quantum observable).

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